

Al-Aqsa University

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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Lecture 2: Linux Command-Line Basics

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Introduction to the Linux Command Line

What is Bash?

- **Definition:** Bash (Bourne Again Shell) is the default command-line shell in Ubuntu and many Linux distributions.
 - **Purpose:**
 - Interprets user commands (e.g., ls, cd) and communicates with the Linux kernel.
 - Executes scripts to automate tasks (e.g., system monitoring in Week 13).
 - **Key Features:**
 - Command execution and history (history command).
 - Scripting support for loops, conditionals (Weeks 5–6).
 - Environment variable management (e.g., \$PATH).
 - **Relevance to Course:**
 - Foundation for command-line tasks in this lecture.
 - Prepares for Bash scripting assignments (Weeks 5, 6, 13).
 - **Example:** Typing ls in Bash lists files by invoking the kernel's file system functions.
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Introduction to the Linux Command Line

- **What is the Command Line?:**
 - A text-based interface (Bash shell) to interact with Linux.
 - Offers precise control over system resources compared to graphical interfaces.
 - Essential for managing intelligent systems (e.g., servers, IoT devices).
- **Accessing the Terminal:**
 - Open with Ctrl+Alt+T in Ubuntu.
 - Or search for "Terminal" in the applications menu.
- **Why Learn Commands?:**

- Direct management of files, processes, and users.
 - Foundation for Bash scripting (Weeks 5–6, 13).
 - Critical for debugging and configuring intelligent systems.
 - **Example:** `ls /etc` lists configuration files, used in system administration.
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Basic Linux Commands: Navigation and Information

- **Core Commands:**
 - `ls`: List directory contents.
 - `ls -l`: Detailed view (permissions, size, date).
 - `ls -a`: Include hidden files (e.g., `.bashrc`).
 - `cd`: Change directory.
 - `cd /home/user`: Go to user's home directory.
 - `cd ..`: Move up one level.
 - `pwd`: Print working directory (e.g., `/home/user`).
 - `man`: Access manual pages (e.g., `man ls` for command help).
 - **Examples:**
 - `ls -la /home`: Lists all files with details.
 - `cd /etc; pwd`: Moves to `/etc` and shows current path.
 - `man cd`: Displays `cd` command documentation.
 - **Tips:**
 - Use Tab for auto-completion (e.g., `cd /ho + Tab`).
 - Press Ctrl+C to stop a running command.
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File Management Commands

- **Creating and Viewing:**
 - `mkdir`: Create a directory (e.g., `mkdir project_folder`).

- touch: Create an empty file (e.g., touch notes.txt).
 - cat: View file contents (e.g., cat notes.txt).
 - **User Information:**
 - whoami: Display current user (e.g., user).
 - **Examples:**
 - mkdir data; touch data/log.txt: Creates directory and file.
 - cat /etc/hostname: Shows system hostname.
 - whoami: Outputs current username (e.g., student).
 - **Use Case:**
 - Create directories for logs or configs in IoT systems.
 - View system files to troubleshoot intelligent devices.
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Advanced Command-Line Features

- **Environment Variables:**
 - echo: Display variable values (e.g., echo \$HOME shows /home/user).
 - echo \$PATH: Lists directories for executable commands.
 - **File Removal:**
 - rm: Delete files or directories.
 - rm file.txt: Deletes a file.
 - rm -r dir: Deletes a directory and its contents.
 - **Caution:** No undo; use carefully.
 - **Command History:**
 - history: Lists previously run commands.
 - !n: Rerun command number n (e.g., !10).
 - history | grep ls: Filters history for ls commands.
 - **Use Case:** Automate repetitive tasks or debug system settings in intelligent systems.
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Expanded Command-Line Examples

- **Scenario 1: System Exploration:**
 - `ls -la /etc`: View all configuration files, including hidden ones.
 - `cat /etc/os-release`: Display Ubuntu version details.
 - `echo $SHELL`: Show current shell (e.g., `/bin/bash`).
 - **Scenario 2: File Management:**
 - `mkdir backups; touch backups/log1.txt backups/log2.txt`: Create directory and files.
 - `ls backups`: List created files.
 - `rm backups/log1.txt`: Delete one file.
 - **Scenario 3: History and Navigation:**
 - `history | grep cd`: Find previous `cd` commands.
 - `cd /var/log; pwd`: Navigate to logs and confirm location.
 - **Why Practice?:** Builds skills for system administration and automation (e.g., Week 13 mini-project).
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Practical Command-Line Scenario

- **Task:** Explore and manage the Ubuntu file system.
 1. Open terminal (`Ctrl+Alt+T`).
 2. Run `pwd` to check your current directory (e.g., `/home/user`).
 3. Use `ls -la` to list all files, including hidden ones.
 4. Run `cd /etc` and use `cat /etc/hostname` to view the hostname.
 5. Create a directory: `mkdir ~/my_project`.
 6. Create files: `touch ~/my_project/note1.txt ~/my_project/note2.txt`.
 7. Run `whoami` to check your username.
 8. Use `echo $PATH` to view executable paths.
 9. Run `history` to view command history.

10.Delete a file: `rm ~/my_project/note2.txt`.

- **Goal:** Master navigation, file management, and system queries.
 - **Record:** Save outputs for the Week 1 assignment.
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Assignment

- **Assignment 1:** Install Ubuntu and Explore Commands
 - **Tasks:**
 1. Install Ubuntu 20.04 LTS in VMware (submit screenshot of desktop).
 2. Run commands: `ls -la`, `cd /etc`, `pwd`, `man ls`, `mkdir test_dir`, `touch test_dir/file.txt`, `cat /etc/hostname`, `whoami`, `echo $HOME`, `echo $PATH`, `rm test_dir/file.txt`, `history`.
 3. Document outputs (e.g., screenshot or copy-paste terminal output) in a PDF or text file.
 4. Submit via Moodle.
 - **Tips:**
 1. Use `man` for command help (e.g., `man mkdir`).
 2. Save terminal output: `history > history.txt`.
 3. Take screenshots with Ubuntu's screenshot tool (PrtSc key).
 - **Deadline:** End of Week 1.
 - **Weight:** 12% of practical grade.
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Next Steps and Resources

- **Complete Tasks:**
 - Practice commands: `ls`, `cd`, `pwd`, `man`, `mkdir`, `touch`, `cat`, `whoami`, `echo`, `rm`, `history`.
 - Experiment with variations (e.g., `ls -l`, `cd ..`, `echo $SHELL`).
- **Review:**
 - Rewatch Lecture 1 and Lecture 2 videos on YouTube/Moodle.
 - Use man pages or Ubuntu forums for command details.

- **Prepare for Week 2:** Linux File System and Permissions.
- **Resources:**
 - Ubuntu Documentation: <https://help.ubuntu.com>
 - Linux Command Cheat Sheet: <https://www.linuxtrainingacademy.com/linux-commands-cheat-sheet/>
 - VMware Support: <https://www.vmware.com/support.html>
 - Moodle: Access lecture videos, ISOs, and assignment submission.